

STEM4U — FTC Engineering Portfolio Template

A section-by-section guide to the portfolio judges actually read · BIOBUZZ 2026–27 *Follow the current Competition Manual for length limits and required elements; this is a starting structure.*

What it is

The Engineering Portfolio is the document judges review **before** they meet your team. It's your team's story — who you are, how you designed the robot, how you run the team, and the impact you make. It feeds nearly every **judged award**, especially **Inspire**. Keep it clear, visual, and honest; judges value real process over polish.

Suggested structure

1. Cover & team summary (½–1 page)

Team number and name, season, school/organization, a photo, and a **3–4 sentence "who we are."** What's your mission and what makes your team distinct?

2. Engineering — the design process (the core)

This is the heart. Show **how you think**, not just the final robot. - **The problem:** what BIOBUZZ asks the robot to do, and your strategy priorities. - **Design iterations:** for each major subsystem (drivetrain, intake/scoring mechanism, etc.) — what you tried, what failed, what you learned, what you changed. **Photos and sketches beat paragraphs.** - **Key decisions & trade-offs:** why this mechanism over the alternatives. - **Testing & data:** how you measured whether it worked (cycle times, reliability).

3. Robot overview (1 page)

Final robot: labeled photo, subsystem summary, and headline specs (drivetrain type, scoring approach, autonomous capability).

4. Team & project management

How the team is organized (roles), how you plan and track work, budget/resources, and your timeline. Judges want to see a **functioning team**, not one hero.

5. Outreach & impact (Inspire-critical)

Who you've reached and the **measurable impact** — events, people, younger students inspired, community partners. Tell one or two real stories, not just a list.

6. Sustainability & the future

How the team funds itself and continues year to year — sponsors, recruiting, mentoring the next class.

Tips that move the needle

- **Show the messy middle.** Judges reward iteration and learning from failure far more than a clean final answer.
- **Photos, captions, diagrams** — a portfolio that's mostly text loses them.
- **Every claim gets evidence** — a number, a photo, or a story.
- **Consistent, readable layout.** One template, clear headers, page numbers.
- **Keep it current.** Update it through the season; the version at your last event should reflect your latest robot.

Pair this with the **Engineering Notebook** (your ongoing, dated log of every meeting). The portfolio is the polished summary; the notebook is the raw record judges may also ask to see.